# Generic HVDC PLUS RMS model for offshore wind farm integration

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# 1 Introduction

A simplified generic RMS model of the symmetrical monopole HVDC PLUS for offshore wind farm integration is included in the simulation software DIgSILENT PowerFactory 2023. The generic control structure is fully accessible for the user without restrictions.

## 1.1 Scope of the model

The scope of the model is described in the following points:

- Valid only for balanced, positive sequence simulations
- Valid only for offshore wind farms connection applications
- Since the focus is mainly on the AC grid on the mainland, the offshore HVDC controller is strongly simplified and is not suitable for detailed offshore-side fault ride through studies.
- The simulation time step can be adjusted from 1 to 10ms. The automatic step size adaption can be used for this model.
- No DC-side faults can be simulated by the model.
- No protection functions are implemented in the model.

# 2 Electrical model

## 2.1 HVDC converters

The HVDC converters are modelled using the Voltage source converter element (ElmVsc) as illustrated in Figure 2-1.

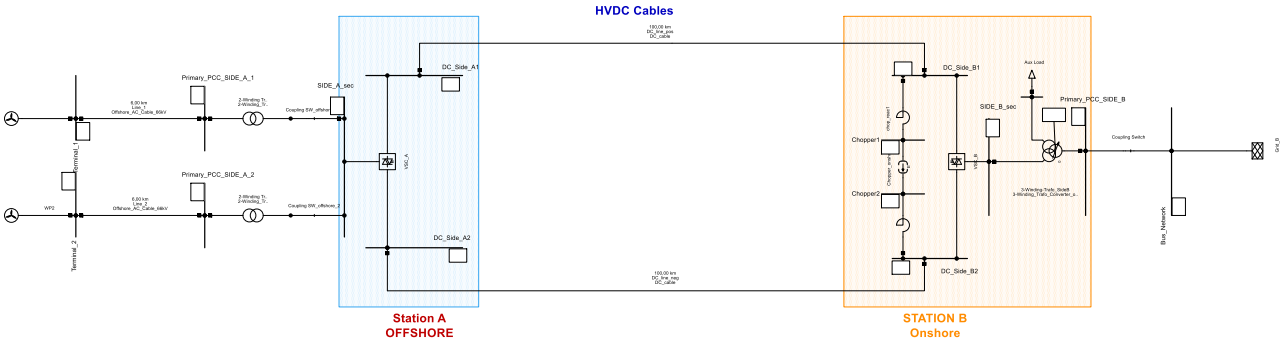



Figure 2-1. Single line diagram of the HVDC PLUS model

### 2.1.1 Offshore HVDC converter

The offshore converter act as a slack for the offshore grid and maintains the AC voltage and frequency. Since the offshore HVDC converter do not directly control the active power, which is delivered by the wind turbines, the load flow control mode Vac-phi is selected for this application.

The dynamic RMS model of the offshore converter receives the magnitude of the pulse-width modulation index (Pm_in) and the frequency setpoint (f0) as input.

### 2.1.2 Onshore HVDC converter

The onshore converter controls the DC voltage and operates in different reactive power modes. The load flow control mode Vdc-Q is used as a default mode. The dynamic RMS model of the onshore converter is modelled as a controlled current source receiving the inputs (id_ref, iq_ ref, cosref, sinref)

### 2.1.3 DC braking chopper

The DC chopper allows decoupling the offshore wind farms from the onshore grid during onshore faults by absorbing the whole energy injected into the HVDC offshore station. The DC chopper is modelled as a controlled DC current source.

### 2.1.4 HVDC Transformers

The HVDC transformers converts the AC voltage level according to the rated voltage at the point of common coupling (PCC). The transformer impedance helps limiting the fault current and provides an extra damping as well for better controllability of the system.

# 3 Generic HVDC PLUS control

## 3.1 Onshore HVDC control

The onshore HVDC control structure is shown in Figure 3-1. The frame includes the following components (slots):

- **Main controller component**: it includes the DC voltage controller, the reactive power controller and the fast voltage control functions.
- **Reactive power control**: the reactive power reference is generated in this component according to the selected control mode.
- **PLL**: phase-locked loop implemented as first order delay.
- **AC voltage measurement**: AC voltage (real and imaginary parts) and frequency.
- **PQ Measurement**: active and reactive power.
- **Chopper control**: generates the DC current reference for the DC chopper.
- **DC Chopper**: modelled as a controlled DC current source with impedance.
- **Converter**: voltage source converter modelled as a controlled AC-current source.

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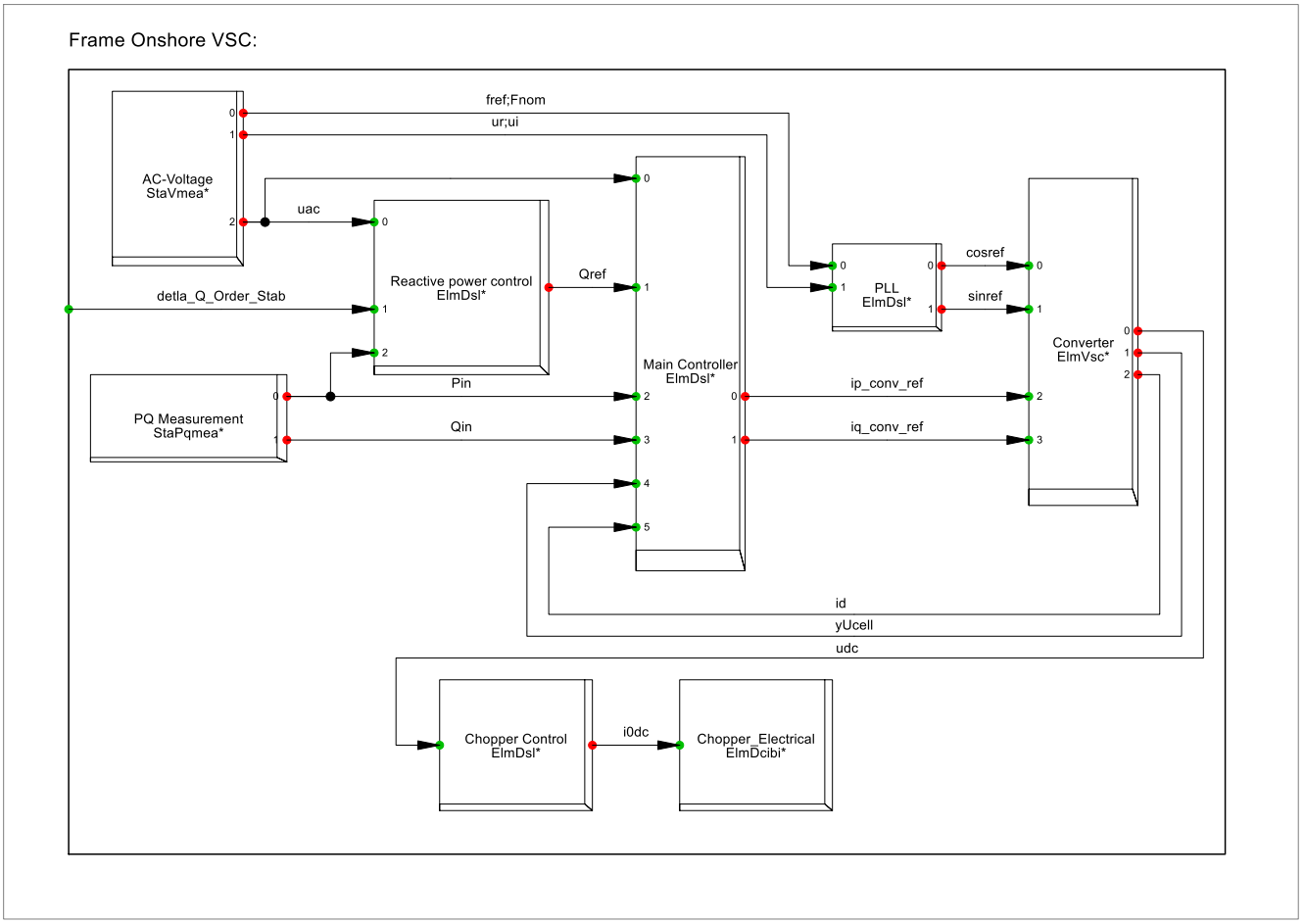



Figure 3-1. Control structure for the onshore HVDC converter

### 3.1.1 Main controller -Onshore

The control structure is illustrated in Figure 3-2. The DC voltage is maintained by adjusting the active current component injected to the AC grid. To ensure a fast and stable dynamic response of the HVDC converter, a feedforward term using the measured active current with first order low pass filter is added to the control path. In addition to the classical PI-controller, an extra dynamic gain is provided by an additional path. This path will be inactive in steady-state due to the considered dead band.

An AC voltage dependent active current limitation ensures reactive current priority during faults. This function is realized by a look-up table. According to several grid codes, the active power (current) must be ramped with a defined ramp rate.

The default values of this controller are listed in Table 3-1 and

Table 3-2.

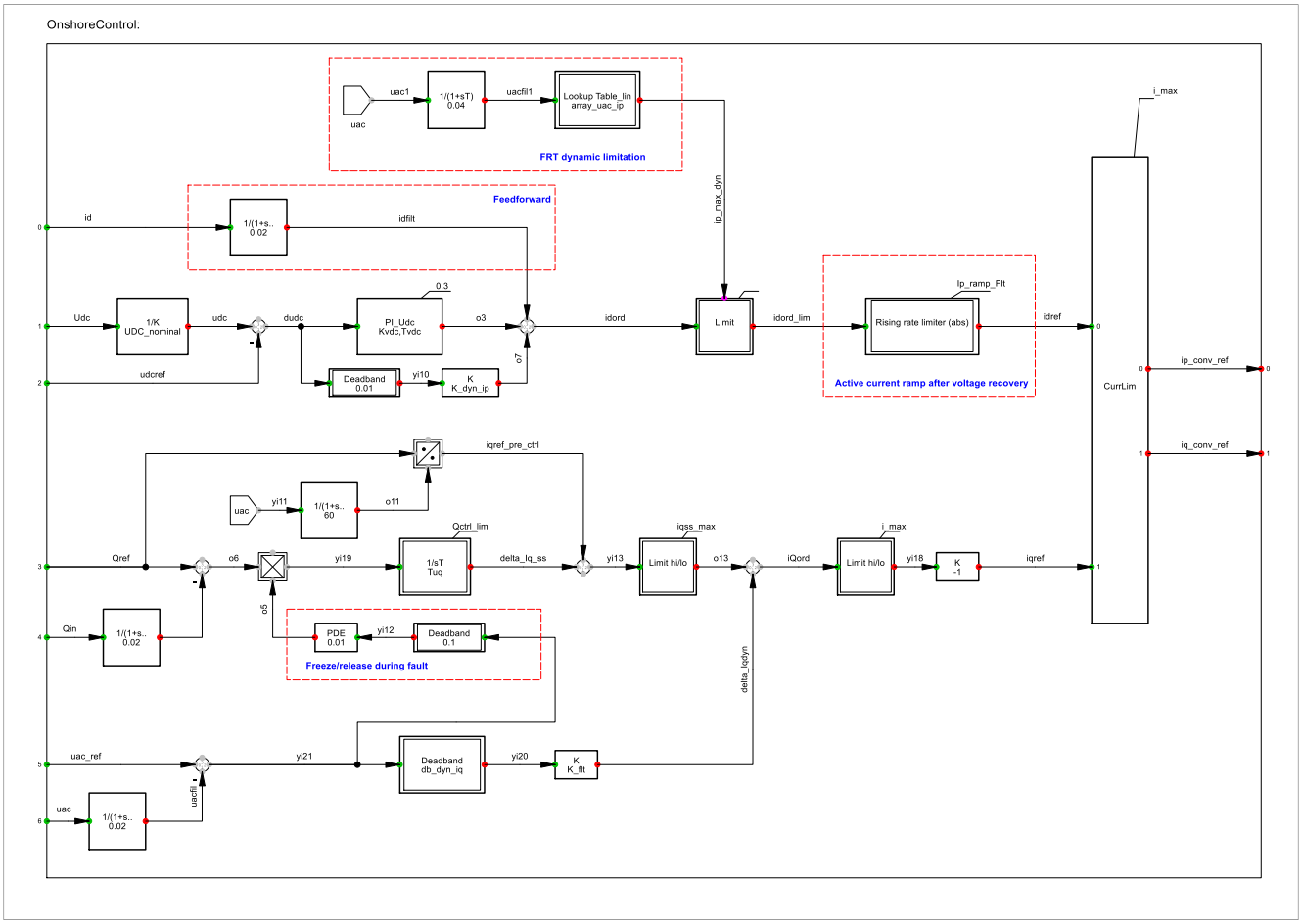



Figure 3-2. Main Controller structure for the onshore HVDC converter

Table 3-1. Parameter list for onshore main controller

| Name | Value | Unit | Description |
|---|---|---|---|
| UDC_nominal | 640 | kV | Nominal DC voltage |
| Tuq | 1 | s | |
| K_flt | 2 | [p.u.] | Reactive current flt gain |
| K_dyn_ip | 6 | [p.u.] | |
| Kvdc | 4 | [p.u.] | Gain of DC PI-Controller |
| Tvdc | 5 | [s] | Integral Time Constant of DC PI-Controller |
| db_dyn_iq | 0.1 | | |
| Ip_ramp_Flt | 9 | [p.u./s] | Ip ramp rate after flt |
| Qctrl_lim | 0.3 | | |
| iqss_max | 0,35 | | |
| i_max | 1.1 | [p.u.] | Current Limit |

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Table 3-2. Look-up table characteristics (AC voltage- active current) for onshore main controller

| Input | Output |
|---|---|
| Uac (p.u.) | ipmax(p.u.) |
| 0 | 0.2 |
| 0,5 | 0.2 |
| 0,85 | 1.1 |
| 2 | 1.1 |
| 2 | 1.1 |
| 2 | 1.1 |

The reactive current order is generated from two functions:

- Slow acting reactive power control: it includes an integrator with a feedforward term.
- Fast local voltage control with deadband and reactive current support gain.

The integrator of the reactive power controller is frozen when the AC voltage is outside the normal operation range (±10%) to ensure stable behaviour during and after fault clearance.

### 3.1.2 Reactive power control

Three control modes are available in the model as provided in Figure 3-6:

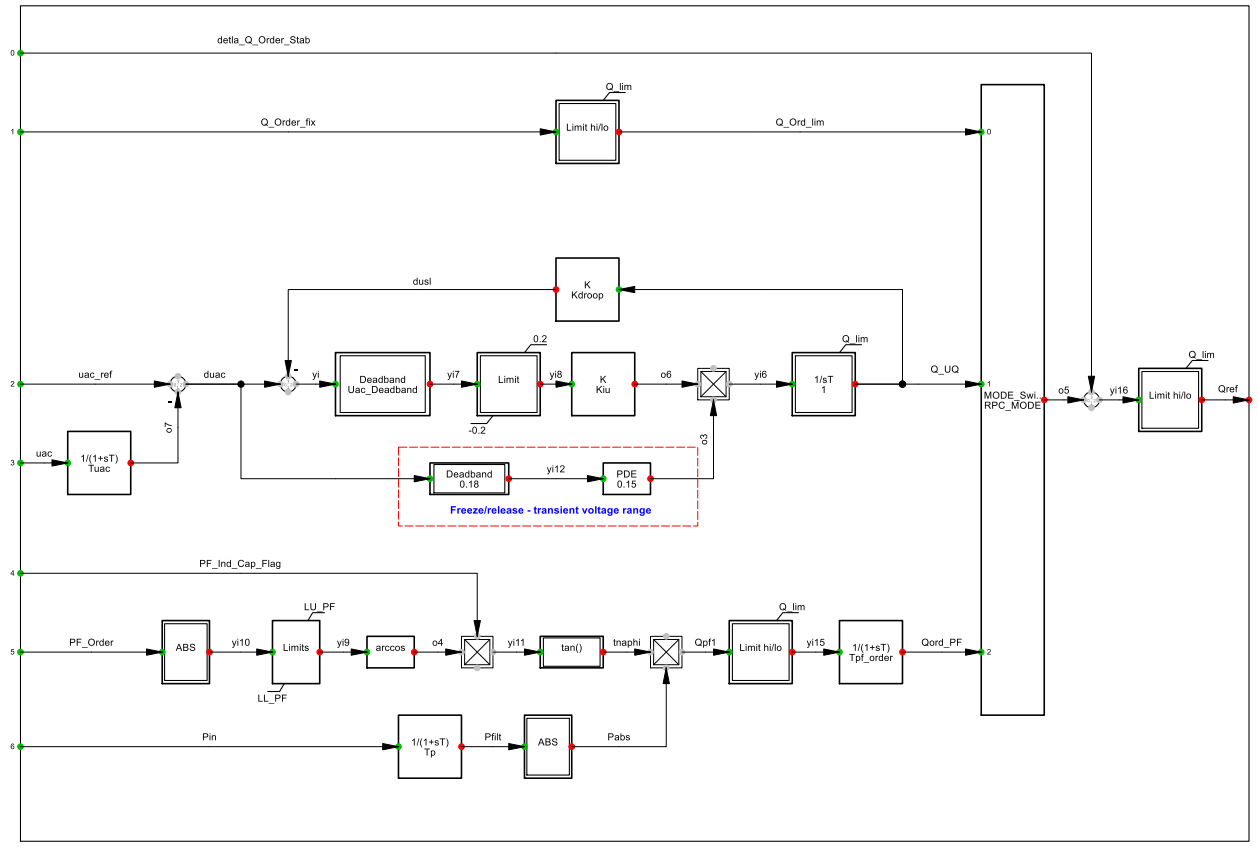
- Voltage droop control mode: the droop is defined by the ratio of the reactive power over the AC voltage in per-units. A dead band for the voltage deviation at the input is additionally provided in the structure. Since the voltage droop control is not a slow acting function, the controller is frozen during transient conditions in which the AC voltage leaves a range of ±18%. In this manner, the voltage support will be solely done by the fast voltage controller in the main controller component. Once the AC voltage recovers, the voltage droop controller will be released after a 150ms delay. This logic is implemented to allow emulating the behaviour of different HVDC systems.
- Power factor control: this a standard control of the power factor based on the measured active power. The measured active power is filtered with a high time constant to ensure a slow action of the control mode.
- Fixed reactive power control: a fixed reactive power control can be given directly in this mode in per-units.

An auxiliary input is added to the model as additional offset to the reactive power reference. This input can be used for further extension of the model with specific stability functions (for example Power damping oscillation).

The default values of this controller are listed in Table 3-3.

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Reactive power control modes:




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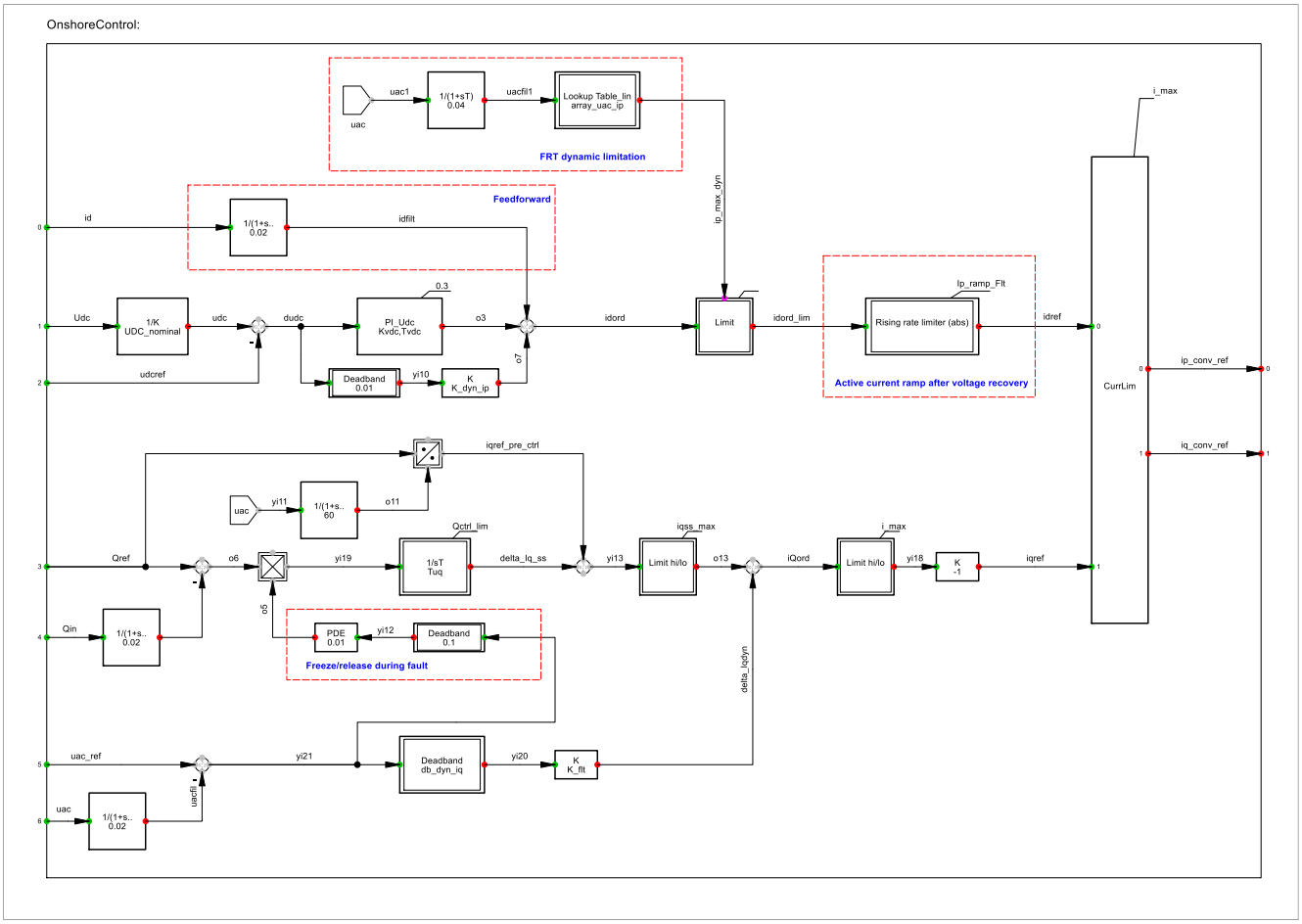



Figure 3-3. reactive power control modes for the onshore HVDC converter

Table 3-3. Parameter list for onshore reactive power controller

| Name | Value | Unit | Description |
|---|---|---|---|
| RPC_MODE | 2 | - | =1: Q control  2:VQ control 3:PF control |
| Tuac | 0,02 | [s] | Filter time constant - measured uac |
| Tp | 0,5 | [s] | Filter time constant - measured P |
| Tpf_order | 1 | [s] | smoothing time constant of PF order |
| Uac_Deadband | 0,05 | [p.u.] | Deadband voltage droop control |
| Kiu | 20 | [p.u.] | proportional gain voltage droop control |
| Kdroop | 0,1 | [p.u.] | inverse Droop in delta_U /delta_Q |
| LL_PF | 0,9 | - | minimum power factor (absolute value) |
| Q_lim | 0,33 | [p.u.] | maximum reactive power |
| LU_PF | 1 | - | maximum power factor (absolute value) |

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### 3.1.3 DC Chopper control

Th DC chopper control is implemented as shown in Figure 3-7. Once the DC voltage exceeds a certain threshold, the DC chopper will become active and generate a DC current reference according to the modified PI control structure. The feedback term using the filtered reference current adds an additional damping to the overall behaviour. A hysteresis is not required in this controller since both the integrator and the final value of the current reference have a lower limit of zero.

The default parameters are listed in Table 3-4.

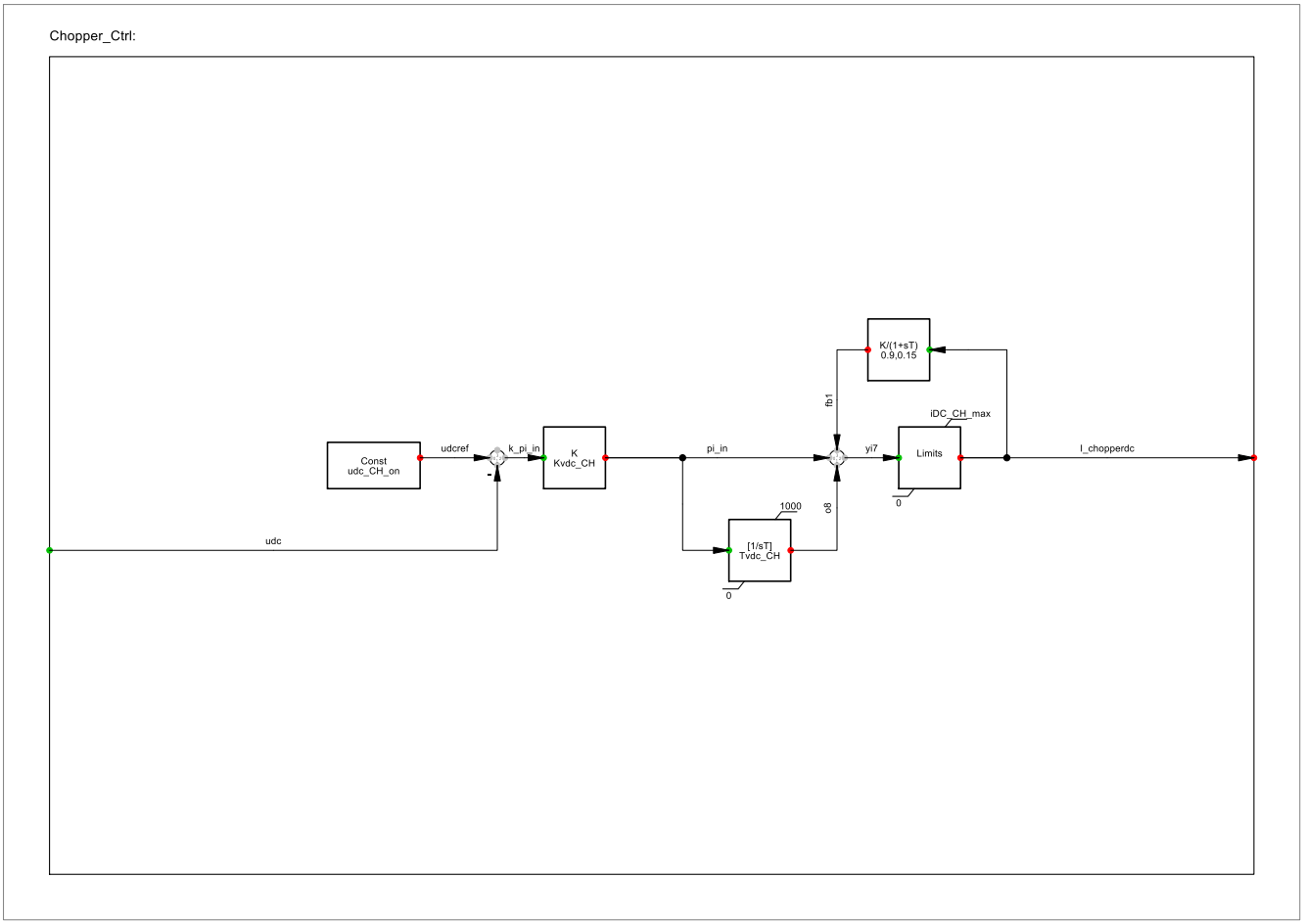



Figure 3-4. DC chopper controller structure for the onshore HVDC converter

Table 3-4. Parameter list for DC chopper control

| Name | Value | Unit | Description |
|---|---|---|---|
| udc_CH_on | 1,035 | [p.u.] | Chopper activation threshold voltage |
| Kvdc_CH | -20 | [p.u.] | Proportional gain chopper control |
| Tvdc_CH | 0,025 | [s] | Integrator time constant - chopper control |
| iDC_CH_max | 1,4 | [p.u.] | Maximum chopper current |

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### 3.1.4 PLL

The PLL structure is shown in Figure 3-5. The real and imaginary parts of the AC voltage are first normalized and then passed through a first-order filter. Since the grid-side voltage is the input for the PLL, the phase shift of the transformer should be considered in the ‘Angle’ function. In addition to this structure, an RMS-specific path is added to consider a realistic dynamic response of the PLL, when the system frequency is changed by the slack. This is achieved by calculating the angle deviation caused by the frequency change in the system.

The default parameters are listed in Table 3-5.

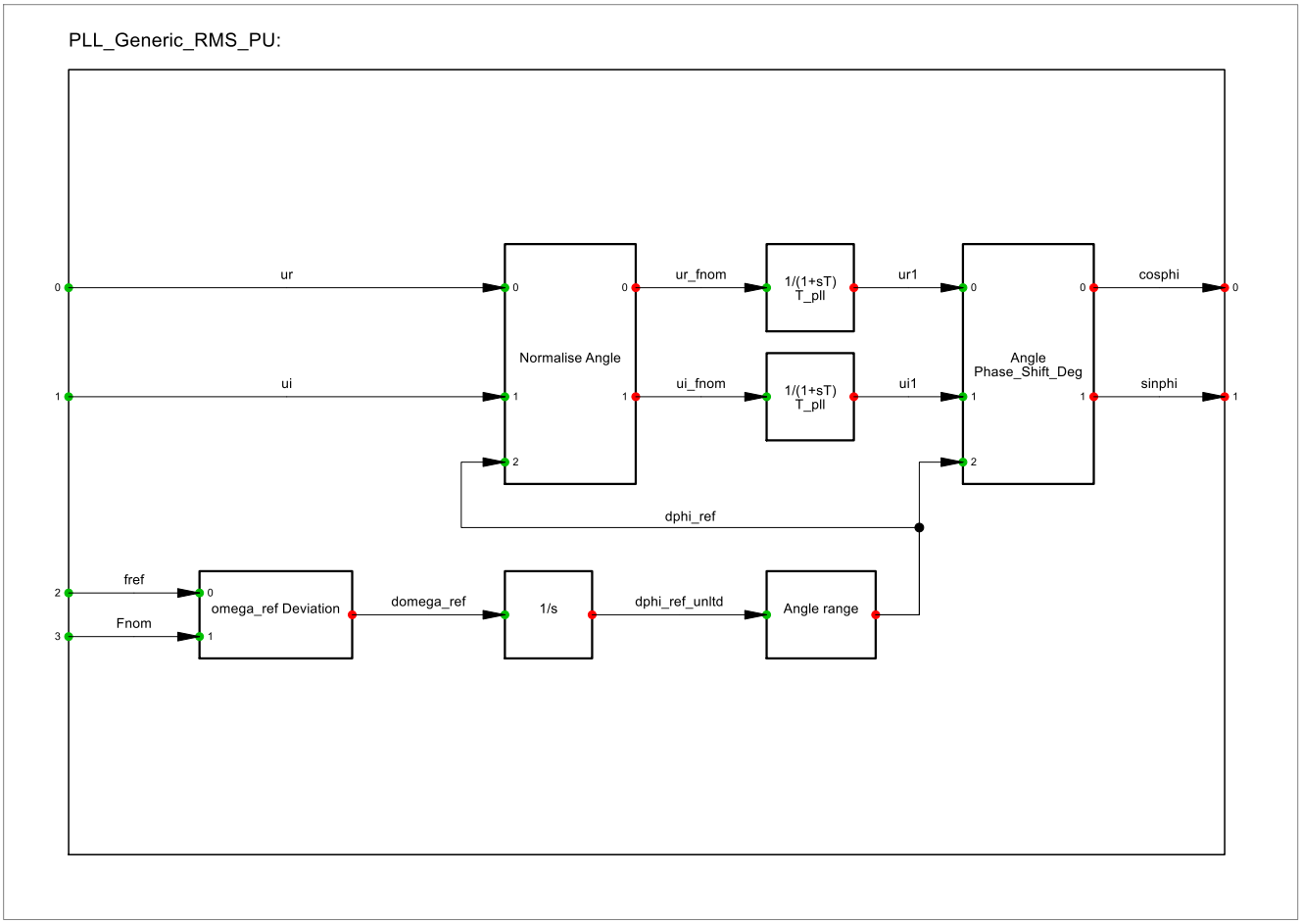



Figure 3-5. PLL structure of the onshore converter

Table 3-5. Parameter list for PLL

| Name | Value | Unit | Description |
|---|---|---|---|
| T_pll | 0,05 | [s] | Filtering time constant for AC voltage |
| Phase_Shift_Deg | -150 | [deg] | Vector group angle of transformer |

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## 3.2 Offshore HVDC control

The onshore HVDC control structure is shown in Figure 3-6. The frame includes the following components (slots):

- **Main controller**: the voltage and frequency references are generated in this component
- **AC voltage measurement**: the magnitude of the AC voltage is measured in per-units.
- **Offshore VSC**: voltage source converter modelled as a controlled AC-voltage source.

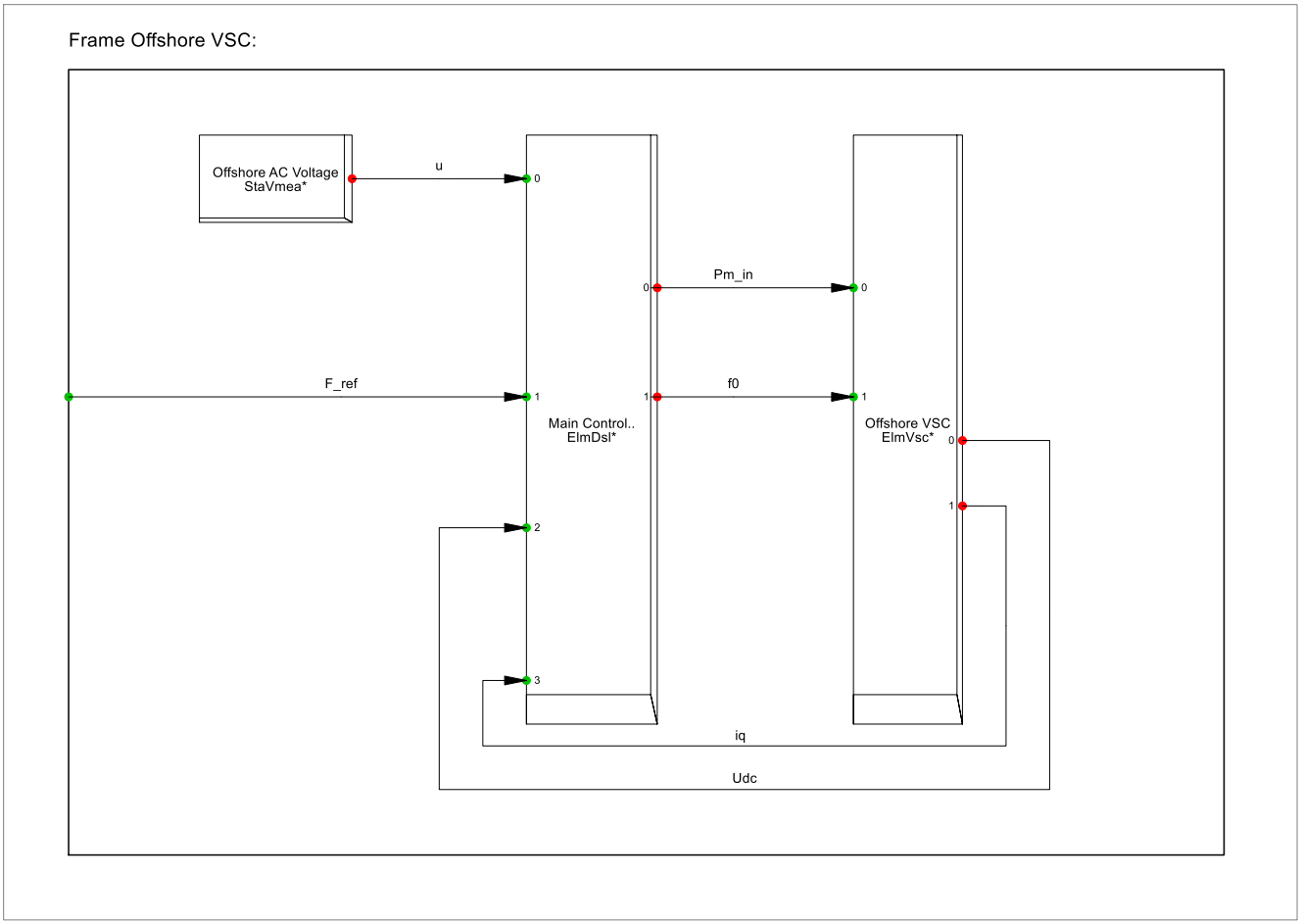



Figure 3-6. Main Controller structure for the onshore HVDC converter

### 3.2.1 Main controller – Offshore

The offshore side controller of this model is very simplified as shown in Figure 3-7. An outer voltage control loop establishes the reactive current reference. In addition to the classical PI-controller, a dynamic proportional gain path is included to ensure optimal voltage support during faults. The output of the second PI controller is then the reference voltage. In the block “Udc_Feedforward” the reference voltage (AC voltage $U$ac) is divided by the measured DC voltage $U$dc and multiplied with a constant which depends on the modulation method (see block: Udc_Feedforward).

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Since the offshore VSC is the slack element in the offshore grid, the AC frequency can be adjusted directly by the frequency setpoint “f0”. To mimic the dynamics of the real controller, a first-order delay is considered for the frequency controller.

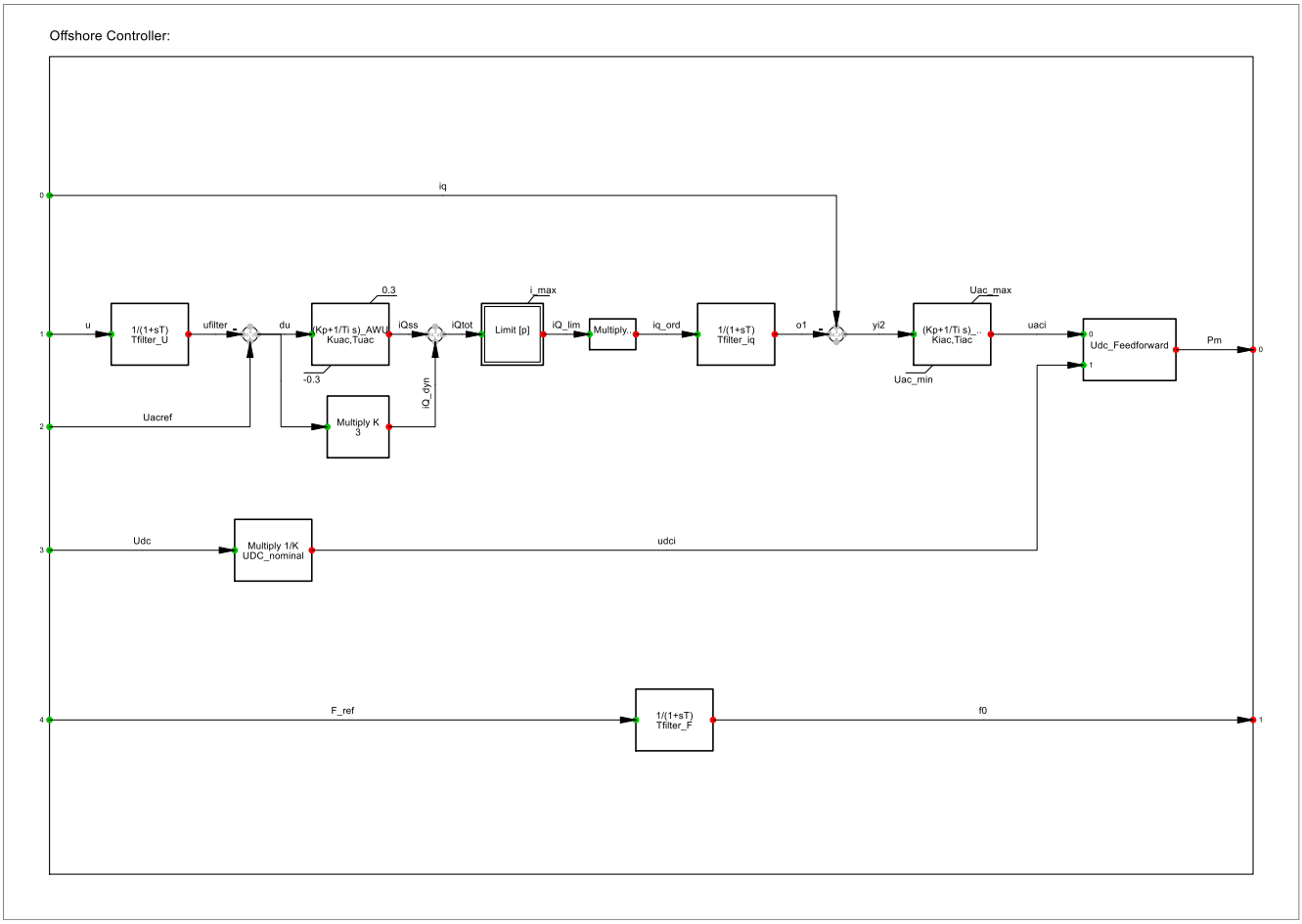



Figure 3-7. Main Controller structure for the onshore HVDC converter

# 4 Model Integration into existing grids

This model can be added as a template to an existing grid using the general template button. The following aspects must be considered when using this model:

- Since control part is in per-unit system, the controller parameters are basically valid for different power and voltage ratings of the HVDC PLUS. If the nominal DC voltage is adjusted in the electrical system, the parameter ‘UDC_nominal’ in the onshore main controller should be adjusted accordingly.
- The power and voltage ratings of the electrical elements must be adjusted manually for different HVDC PLUS ratings.

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- Since the Power measurement at the onshore point of common coupling is defined based on fixed power rating input (see Figure 4-1), this value must be adjusted manually for different power rating

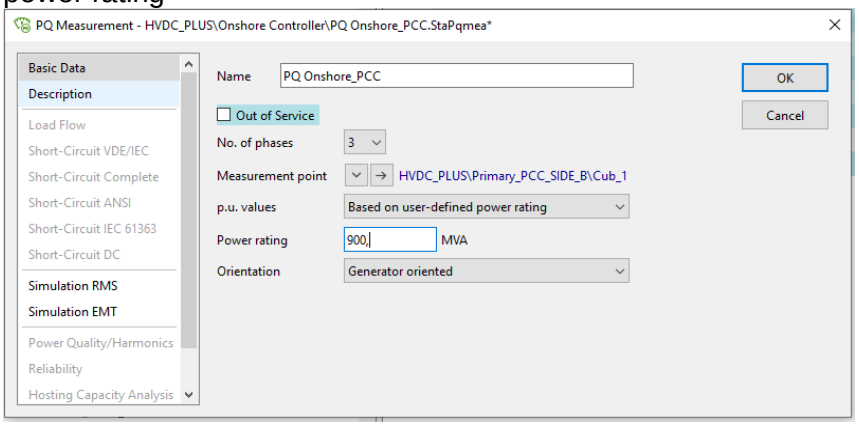



Figure 4-1. PQ-measurement block at the HVDC onshore converter station

- At the offshore side, the HVDC PLUS converter does not use any measurement at the 66KV as an input for the converter control. The AC voltage is controlled only the converter side bus (SIDE_A_sec). The offshore transformers can therefore have different topologies (3-winding transformers) or different number of feeders without the need to adjust any control signals.

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